Medical Robotics

Prof. Marilena Vendittelli
Sapienza Università di Roma

Lecture 3
Classification of Surgical Systems

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1. Why Classification Matters

Classification of surgical robotic systems has itself been a topic of active research. In the early years of the field — late 1990s, early 2000s — one of the central open questions was exactly how to classify these systems: should a given device be considered passive, or if it has some degree of autonomy, what *kind* of autonomy? The shift from the binary "passive vs. active" language of the earliest literature toward the more nuanced, multi-dimensional frameworks we discuss here reflects the growing maturity of the field.

Classification serves three main purposes:

- **Structured understanding**: it provides a structured picture of the systems produced for medical/surgical application, making it easier to identify, compare, and evaluate them.
- **Intended use identification**: it helps to identify the intended use of a specific system and to determine the *essential performance* — the minimum functional requirements the device must maintain for patient safety — that must be demonstrated for certification.
- **Technology analysis**: for the engineer, it is the lens through which to analyse which methods and technologies are relevant, which is the primary purpose of classification in this course.

> **No single scheme is complete.** Each classification scheme described in this lecture reflects the perspective and goals of its authors. They are *complementary*, not competing. A real system is typically best described by more than one scheme simultaneously. The classification used throughout this course — based on *robotic task and domain of use* — is chosen because it is the most useful lens for the engineering analysis of design methodologies.

2. Classification 1: Troccaz (2001) — Interaction Mode

The first widely cited classification in surgical robotics was proposed by Jocelyne Troccaz who designed important surgical systems — among them PADyC, discussed below. At the time of her classification, the central question being debated was not so much "level of autonomy" as the more binary question: should a surgical system be passive, or should it be active, and if active in what way? Troccaz introduced the concept of *interactive systems* as a third category distinct from both fully passive and fully active. This concept of shared physical guidance — the machine and the surgeon both involved in controlling the instrument — is what we now call **shared control**, a paradigm that today is widely used not only in surgical robotics but also in industrial robotics, where it integrates the flexibility of human intelligence and experience with the accuracy of the robot.

> **Passive systems**: unactuated, manually driven. No motors; joints are only instrumented with sensors. The surgeon drives all motion; the system provides information.
>
> **Active systems**: autonomously perform the pre-planned procedure under human supervision. The robot executes the motion; the human monitors and can stop the procedure. What we today call "task-level autonomy": the robot is not truly autonomous — it executes a plan provided by the surgeon — but it does so without real-time motor input from the human.
>
> **Interactive systems**: provide physical guidance to the surgeon through control and/or specific hardware. The surgeon and robot share physical control. Two sub-classes:
>
> - *Semi-active*: the robot pre-positions a *mechanical* guide; the constraint is fixed once the

robot locks.

- *Synergistic*: the constraints are *programmable*, not mechanically fixed. Today we would say: shared control using virtual fixtures.

Teleoperated systems: the surgeon operates a master device; the robot replicates those motions at the patient side.

2.1. Passive Systems

Passive systems are easier to understand than they might first appear, because we have already encountered several of them in the systems discussed in Lectures 1 and 2. Consider the **master manipulator of the Da Vinci system** (prior to the Da Vinci 5): the device that the surgeon holds and moves is an articulated arm with no motors — its joints are only instrumented with encoders. It is a passive manipulator that measures the surgeon's hand motion and maps it to the slave robot. This is a passive system in Troccaz's sense: it contains no actuation; its only function is to sense and transmit information.

Similarly, an **optical tracking system** using fiducial markers — such as the infrared camera systems used in surgical navigation — is a passive system: it provides information about the position of the patient and tools but executes no motion itself. Its limitation is that it requires a free line-of-sight between the camera and the markers; if something passes between them, tracking is lost.

2.1.1. The Viewing Wand (ISG Technologies)

The Viewing Wand is an articulated arm — similar in form to a lightweight robot arm — whose joints are instrumented with encoders but have no motors. The surgeon manually moves a probe at the tip; the encoders, through forward kinematics, allow the system to compute the 3D position and orientation of the probe in the operating room.

One important application is neurosurgical navigation for electrode implantation (for treatment of epilepsy, severe depression, or brain tumour biopsy). The workflow is: preoperative CT imaging $\rightarrow$ 3D reconstruction of the patient's anatomy $\rightarrow$ surgical planning. Once the patient enters the operating room, the critical step is **registration**: aligning the preoperative plan (in imaging coordinates) with the actual patient position (in OR coordinates). The Viewing Wand achieves this by touching known anatomical landmarks or fiducial markers, establishing the coordinate frame correspondence. Once registered, the arm allows the surgeon to navigate — to see where the probe tip is, relative to the preoperative anatomy, in real time.

Advantages	**Drawbacks**
Stability of positioning if the arm has brakes	Cumbersome; constrains the surgeon's movement range
No need for a free line-of-sight (unlike optical tracking)	Limited functionality — purely informative

Passive systems in navigation. Navigation systems that track tool position relative to the preoperative image — BrainLab, Stryker Navigation — are arguably the most widely deployed class of "surgical robotics" in current clinical practice. They are passive in Troccaz's sense: they generate no motion; they provide information. Yet their clinical impact in neurosurgery, spine surgery, and orthopedics is immense.

2.2. Active Systems

Active systems autonomously execute a pre-planned procedure. The surgeon does not guide the robot's motion in real time; they approve the plan beforehand and supervise execution, with the ability to halt at any moment. The universal workflow is: preoperative imaging → patient-specific model → surgical planning → intraoperative registration → autonomous robotic execution → verification.

Active systems are indicated for the following tasks:

- **Machining of rigid surfaces**: ROBODOC, CASPAR (bone milling for arthroplasty).
- **Carrying/holding heavy or precisely aimed tools**: CyberKnife (linear accelerator on a robot arm).
- **Controlled physical interaction**: TMS Robot (magnetic coil held in physical contact with the patient at prescribed position with limited force), MELODY (active force control of ultrasound probe at patient side). Note that MELODY belongs to two categories simultaneously — it is also a teleoperated system, because the probe position is commanded remotely — but the slave manipulator has its own active force controller. This is an example of **shared control**: part of the task is executed under the direction of the remote physician, part is autonomously completed by the robot controller.
- **Intra-body tasks**: active endoscopic capsules (pills that move under their own control or under external magnetic fields, rather than being passively transported by peristalsis).
- **Moving targets**: CyberKnife with real-time patient motion tracking. The patient wears a sensor jacket that detects respiratory motion; the robot synchronises the beam direction with that motion continuously throughout the treatment.

2.2.1. ROBODOC (Think Surgical / ISS, 1992)

ROBODOC was the first FDA-cleared surgical robot (1992) and the first purpose-built active surgical system. It performs total hip arthroplasty (THA) and total knee arthroplasty (TKA) — procedures requiring precise milling of the femoral canal or tibial/femoral bone surfaces to accept a prosthetic implant.

Kinematic architecture: SCARA robot with a 2-DOF wrist. The SCARA architecture was chosen because it provides high stiffness in the horizontal plane, which is where bone-cutting forces are applied.

Key specifications:

- Accuracy < 1 mm, achievable because the patient is rigidly fixtured and registration uses implanted titanium bone pins.
- 6-DOF force/torque sensor at the end-effector for real-time force monitoring and safety stop.
- **Orthodoc planning software**: the surgeon segments the CT scan, places the virtual implant in the 3D model, and defines the milling trajectory. ROBODOC executes this trajectory autonomously.

Surgical workflow:

1. Preoperative CT scan; 3D bone model generation with Orthodoc.
2. Titanium fiducial pins implanted in the bone *before* the main operation (a separate minor surgical step).

3. Intraoperative registration: the robot's probe touches the pin locations to establish the coordinate frame correspondence.

4. Autonomous bone milling with continuous force monitoring.

> **Why titanium pins, and why is registration critical?** ROBODOC uses implanted titanium pins as fiducial points for registration. These pins are rigidly fixed to bone and visible in imaging, so their coordinates in the image frame are known precisely. By touching them with the robot in the OR, the transformation between the image frame and the robot frame is established. Sub-millimetre accuracy of the final milling is only possible if this registration is also sub-millimetre accurate. The requirement for a separate pin-implantation procedure is a significant workflow burden and was one of the main limitations of ROBODOC's clinical adoption. Later systems (MAKO, TSolution One) moved to intraoperative registration using probes touched to anatomical landmarks, eliminating the pre-operative pin step.
>
> ROBODOC evolved into the **TSolution One** (Think Surgical, USA), which remains in clinical use today.

2.2.2. CASPAR (URS Ortho, Germany)

CASPAR pursues the same clinical goals as ROBODOC — autonomous CT-planned bone milling for arthroplasty — with a different kinematic architecture: a standard **6-DOF anthropomorphic** (industrial-type) robot arm rather than a SCARA. The anthropomorphic configuration offers a larger workspace and greater flexibility of tool approach angle, at the cost of less inherent stiffness.

CASPAR also improved on ROBODOC's registration by allowing **intraoperative patient motion tracking**: if the patient moves during the procedure, the registration can be updated. This addresses one of the main risks of fully autonomous bone milling.

CASPAR was discontinued around 2002 following clinical trial complications, primarily related to registration accuracy issues — demonstrating that the overall system accuracy depends critically on the registration step, not only on the robot's intrinsic repeatability.

2.2.3. CyberKnife (Accuray)

CyberKnife was motivated by a specific clinical problem in stereotactic neurosurgery: the invasive rigid head frame that must be bolted to the skull to immobilise the patient and provide a reference frame for the radiation beam. This frame causes patient discomfort, limits treatment to a single session, and cannot be used for extracranial tumours (because the frame can only be bolted to the skull).

Technology: a 6-DOF KUKA robot arm carrying a 6 MeV linear accelerator. The robot's large workspace and dexterity allow the beam to be aimed from hundreds of directions around the patient. Key features:

- **Frameless**: no stereotactic head frame required. Position is tracked intraoperatively using X-ray fluoroscopy and/or implanted fiducial markers.
- **Real-time motion tracking**: the robot monitors tumour position continuously and adjusts beam direction to compensate for breathing, cardiac motion, and involuntary movement. The patient wears a sensor jacket that monitors respiratory motion, and the system synchronises beam delivery with the respiration cycle.
- **Non-isocentric treatment**: beams from independent directions converge only at the tumour volume, delivering lethal dose to the tumour while sparing surrounding tissue.

- Sub-millimetre targeting accuracy.

> **CyberKnife in Troccaz's classification.** CyberKnife is unambiguously active: it autonomously executes a pre-planned treatment programme. The human role during treatment is supervisory — monitoring vital signs and having the ability to halt the system. The robot generates and executes its own beam-direction trajectory without real-time human motor input. This is the correct meaning of "active" or "task-level autonomy": not unsupervised, but not requiring real-time human motor guidance either.

2.3. Interactive Systems

Interactive systems are designed around **shared control**: the robot constrains or guides the surgeon's motion without taking over entirely. The surgeon remains physically in the loop at all times, retaining direct haptic feedback. This architecture offers important safety and psychological acceptance advantages over fully active systems: the surgeon always feels that they are in control, even though the robot is constraining their motion.

2.3.1. Semi-active Systems: NeuroMate

In a semi-active system, the robot pre-positions a **mechanical guide**; the surgeon then acts within the constraint that guide imposes. The constraint is physical and fixed: once the robot locks its joints, it holds completely still. The semi-active concept is closely related to the PUMA 560 experiment of 1985 (Lecture 2): the PUMA was used to position a needle guide on the skull to target a brain tumour, after which the surgeon inserted the needle manually. That was among the first examples of a semi-active system.

NeuroMate (Renishaw Mayfield, UK) formalises this approach as a purpose-built clinical system. It constrains the surgeon to specific motion primitives:

- **Linear motions**: e.g., needle or electrode insertion along a pre-planned axis.
- **Planar motions**: e.g., an osteotomy (bone cut) constrained to a planned plane.
- **Conical motions**: e.g., the laparoscopic trocar pivot (RCM constraint).

Specifications: 5-DOF arm; FDA-cleared for neurosurgery; frameless or frame-based stereotaxis; image-guided trajectory targeting. Because the constraint is purely mechanical (locked joints), no powered motion can occur during the intervention — an intrinsic safety property.

2.3.2. Synergistic Systems: PADyC

Synergistic systems generalise the semi-active concept: constraints are **programmable in software** rather than fixed mechanically. This means the admissible motion region can change dynamically during the procedure as the clinical context evolves. In today's terminology, these are **virtual fixtures** implemented through control; the concept introduced by Troccaz as "synergistic" is essentially what we now call shared control via virtual fixtures.

PADyC (Passive Arm with Dynamic Constraints, TIMC Grenoble) implements programmable constraints using a novel actuation mechanism: *freewheels*. At each joint, two freewheels are mounted in opposite directions, each driven by an independent motor. The key mechanical insight is this: a freewheel transmits torque in only one rotational direction. When the internal motor axis is held still, the external ring (driven by the surgeon's hand) can rotate freely in one direction but is blocked in the other. By combining two freewheels in opposite orientations and regulating the speed of each motor axis, the system creates a *window of admissible velocities* — a range $[\omega_i^-, \omega_i^+]$ within which the surgeon can move the joint freely, but outside which motion is mechanically blocked.

Mode	**Condition**	**Permitted motion**
F_1	$\omega_i^- = 0,\ \omega_i^+ \neq 0$	Positive direction only
F_2	$\omega_i^- \neq 0,\ \omega_i^+ = 0$	Negative direction only
F_3	$\omega_i^- \neq 0,\ \omega_i^+ \neq 0$	Both directions
F_4	$\omega_i^- = 0,\ \omega_i^+ = 0$	Locked (no motion)
Admissible velocity: $\omega_i^- \leq \omega_i \leq \omega_i^+$		

The admissible region is planned *in task space* (Cartesian coordinates of the surgical tool) based on preoperative imaging. But the freewheel limits must be expressed in *joint space*. This is the fundamental computational challenge of PADyC: mapping task-space constraints to joint-space velocity windows, in real time, for multiple control points along the surgical tool. The overall Window of Admissible Configurations (WAC) is the intersection of individual WACs computed for each control point:

$$WAC(t) = \bigcap_{j=1}^{k} WAC_j(t)$$

PADyC operating modes:

- **Free mode**: no constraint; the system records the tool trajectory (used for registration and free exploration).
- **Position mode**: guides the user toward a predefined target pose (e.g., correct prosthesis placement).
- **Region mode**: the tool is free within a defined spatial region but cannot leave it. Two variants: *keep inside* (e.g., tumour resection volume; avoidance of critical structures) and *keep outside* (safety margin around vessels or nerves).
- **Trajectory mode**: constrains the tool to follow a predefined path within a given accuracy tolerance (e.g., a biopsy trajectory).
- **Specialised modes**: linear, planar, and conical motion primitives.

> **Why PADyC had limited clinical success.** Despite its elegant mechanical design and intrinsic safety (the constraint is enforced by physics, not software), PADyC had limited adoption for two reasons. First, mapping complex surgical gestures from task space to the joint configuration space of the manipulator is computationally demanding and numerically ill-conditioned near kinematic singularities. Second, many practical surgical gestures are difficult to fully specify as velocity-window constraints in joint space. The system was clever but proved too complex to deploy reliably across the variety of real surgical scenarios. It is no longer commercially available, but it was intellectually important and its conceptual descendants — virtual fixtures implemented via impedance control — remain highly relevant.

2.3.3. Synergistic Systems: Acrobot (Imperial College London)

The **Acrobot** (Active Constraint ROBOT) implements shared control using **impedance control** rather than hardware freewheels. This is a purely software approach: the constraint boundary is defined in the controller, not in any physical mechanism.

Operating principle:

- **Inside the safe zone**: the robot is *transparent* — it offers essentially zero resistance, and the surgeon feels as if operating without a robot. The impedance controller is set to very low stiffness.

- **At the boundary**: the controller computes the distance from the tool tip to the constraint surface and applies a resistive force proportional to penetration depth — a high-stiffness virtual wall. The surgeon feels increasing resistance as they approach the boundary and cannot easily push through.
- The surgeon retains full haptic feedback throughout, including the feel of the cutting tool interacting with bone.

This will be studied in a dedicated lecture on impedance control and virtual fixtures. The Acrobot evolved into the Sculptor system and is now the **MAKO** system (Stryker), currently the world's leading hands-on orthopedic robot.

> **PADyC vs. Acrobot: two implementations of synergistic control.** Both are synergistic (shared control) but implement the constraint very differently:
>
> - **PADyC**: passive arm + hardware freewheels + velocity windows. Constraint enforced by physics. Cannot generate powered motion ⇒ intrinsically safe. But computationally demanding to map task constraints to joint space in real time.
> - **Acrobot/MAKO**: active arm + impedance control + virtual fixture. Constraint enforced through control. More flexible, complex constraint geometries defined in the task space. Requires reliable real-time computing.

2.4. Teleoperated Systems

In teleoperated systems, the surgeon operates a master device; a slave robot at the patient side replicates those motions. The surgeon is spatially separated from the patient. Key engineering benefits: motion scaling, tremor filtering, and ergonomic operation. The teleoperated paradigm is the gold standard of minimally invasive surgery today.

Until very recently, the slave robot in a teleoperated surgical system was completely passive in its decision-making: it executed the surgeon's commands with no independent reasoning. Today, even teleoperated systems are beginning to incorporate some degree of AI-assisted guidance, supporting the surgeon's reasoning and decision-making in real time. This is the direction toward increasing LoA within the teleoperated paradigm.

2.4.1. Da Vinci (Intuitive Surgical, 1999–present)

- Three subsystems: Surgeon Console, Patient Cart (3–4 arms), Vision Cart.
- EndoWrist instruments: 7 DOF at the tip, exceeding human wrist dexterity.
- 3D HD stereo vision (10× magnification); motion scaling and tremor filtering.
- **Da Vinci 5 (2024)**: first generation to incorporate **force feedback capability** — a long-awaited feature addressing one of the most discussed limitations of all previous generations.

2.4.2. Zeus (Computer Motion, 1998–2003)

Zeus competed directly with Da Vinci, using three robot arms mounted to the OR table and integrating the AESOP voice-controlled camera arm. It was used in the Lindbergh Operation (2001, first transatlantic surgery). Computer Motion was acquired by Intuitive Surgical in 2003, ending the direct competition.

3. Classification 2: Taylor (2003) — Computer-Integrated Surgery

R. H. Taylor of Johns Hopkins University is considered by many to be the most important figure in the founding of modern surgical robotics. His 2003 classification frames surgical robots within the broader context of **Computer-Integrated Surgery (CIS)** — a concept directly modelled on Computer-Integrated Manufacturing (CIM), which had been a major paradigm in industrial robotics since the 1980s. The translation of this concept from manufacturing to surgery was not straightforward, but it was prescient: it correctly identified that a surgical robot cannot be understood as an isolated device. It is a component of a larger information-processing and physical-execution system.

> **The fundamental insight.** In executing a robotic surgery, an enormous amount of data is generated and recorded that was impossible to capture in manual surgery: tool positions, forces, timings, deviations from plan, imaging data, outcomes. A surgical robotic system must be designed from the outset to capture, process, and feed back this data into the clinical workflow — preoperative planning, intraoperative execution, postoperative assessment, and future planning. This is the CIS vision. Taylor was right, and it is increasingly recognised that AI will exploit exactly this data stream to progressively increase the autonomy and effectiveness of surgical robotic systems.

3.1. The CIS Framework

The CIS lifecycle has three phases:

- **Preoperative**: computer-assisted planning based on a patient-specific model (from imaging) and an atlas of general anatomical/procedural knowledge. No robot here, but computer science — and increasingly AI for image segmentation and registration.
- **Intraoperative**: the plan is transferred to the robot. As in any feedback control system, the open-loop execution of the plan will not yield the desired result exactly — tissue moves, the patient shifts, anatomy deforms. Intraoperative imaging (e.g., C-arm fluoroscopy for real-time X-ray) provides the feedback signal: the model and plan are updated, and execution is corrected. This feedback loop is the CIS equivalent of closed-loop control.
- **Postoperative**: computer-assisted assessment and outcome documentation. In robotic surgery, the volume and richness of recorded data far exceeds what is available from manual surgery. This data feeds back into the atlas, improving future planning and training.

Within this framework, surgical robots belong to one of two sub-categories: **surgical CAD/CAM** or **surgical assistants**.

3.2. Surgical CAD/CAM

The analogy to mechanical engineering is direct: CAD defines the target geometry; CAM translates it into machine tool trajectories; the machine executes them. In surgical CAD/CAM: preoperative imaging and planning define the target (the correct bone resection, the tumour volume to be irradiated); the planning software generates the robot trajectory; the robot executes it. **Registration plays the role of workpiece fixturing**: it establishes the correspondence between the planned geometry (in imaging coordinates) and the physical patient (in robot coordinates). Errors in registration directly degrade execution accuracy, regardless of the robot's intrinsic precision.

Pipeline: patient modelling → planning → registration → execution → follow-up.

Examples:

- **Robot-assisted execution**: ROBODOC (bone milling); NeuroMate (needle guide positioning for neurosurgery).
- **Navigation-assisted execution**: CyberKnife (multimodal image-guided beam delivery); augmented reality systems (preoperative image overlay on the intraoperative view).

3.3. Surgical Assistants

Surgical assistants work *alongside* the surgeon rather than replacing the surgeon's motor action. Two sub-classes:

- **Surgical extenders**: improve or extend the surgeon's physical abilities. Examples: teleoperated systems (Da Vinci, Zeus); microsurgical systems; cooperative/synergistic systems.
- **Auxiliary surgical supports**: work side-by-side with the surgeon to hold or position instruments. Examples: AESOP (endoscope holding), active retractors, systems for intraluminal applications.

> **In what sense is Da Vinci a "surgical extender"?** It extends the surgeon's capabilities in several ways simultaneously: it allows access to internal areas through very small wounds (reducing harm to the patient); it scales down and filters the surgeon's gestures (improving precision); it provides 3D magnified vision (improving perception). The system does not replace the surgeon's decision-making — every motion originates from the surgeon — but it extends what the surgeon's hands can physically accomplish. This is the correct way to think about it: not a replacement, but an extension.

3.3.1. AESOP (Automated Endoscopic System for Optimal Positioning)

AESOP exemplifies the auxiliary surgical support category. It is a **voice-controlled 7-DOF robotic arm** that holds and repositions the endoscope during laparoscopic surgery. In MIS, having a third "hand" to hold the camera while both the surgeon's hands are occupied with instruments is one of the persistent practical difficulties. AESOP resolves this by placing camera control under the surgeon's direct verbal command.

Clinical advantages:

- **Steady camera**: a human assistant introduces tremor, drift, and fatigue, particularly over long procedures.
- **Both hands free**: the surgeon can operate two instruments simultaneously without diverting attention to camera repositioning.
- **Reduced OR staffing**: the camera-holder role is eliminated.
- **No miscommunication**: the surgeon issues direct voice commands rather than verbal instructions to a human assistant. In practice, miscommunication about camera direction ("move left a little — the other left") is a real source of OR inefficiency and occasionally of errors.

Voice interface: commands include *"scope in"*, *"scope out"*, *"move up/down/left/right"*, with the system trained on the individual surgeon's voice patterns. AESOP was FDA-cleared in 1993 and was later integrated as the camera arm of the Zeus system.

4. Classification 3: Joskowicz (2005) — Type of Support

Joskowicz's classification was introduced by the researcher who developed the MARS bone-mounted robot (Mazor). It takes a **clinical and practical** perspective, spanning the full range from completely passive mechanical aids to active robots, and explicitly including non-robotic solutions.

The hierarchy of support devices:

1. Passive mechanisms (adjustable frames, arms, supports).
2. Individual templates (patient-specific passive guides, e.g., 3D-printed).
3. Intraoperative imaging and navigation.
4. Robotics: floor/bed-mounted; patient-mounted.

4.1. Pedicle Screw Insertion: A Worked Example

Pedicle screws are inserted through the bony pedicle of the vertebra to provide anchor points for spinal fixation constructs. The pedicle is narrow (5–7 mm in the lumbar spine, less in the thoracic spine), and the spinal cord and nerve roots are immediately adjacent. Indication for fixation includes vertebral fracture, degenerative disease, spinal tumour, and scoliosis correction. Statistics from 10–15 years ago: **10% to 40% of screws were ill-placed** — a clinically unacceptable rate, and the motivation for developing assistance systems. The imaging used was CT (tomographic), providing 2D sections from different angles. The surgeon builds a mental 3D model of the patient anatomy and executes the surgery based on that mental model. Experience and skill are critically important, which explains the wide variability in the accuracy statistics.

Individual template (passive device): Based on the preoperative CT and 3D digital model of the patient, a passive guide (today often 3D printed) is manufactured to fit exactly onto the posterior surface of the target vertebra. A channel aligned with the planned screw trajectory provides a rigid mechanical guide for the drill. This works like a ruler for drawing: completely passive, but mechanically constraining.

- Advantages: inexpensive; rigid mechanical support; fully customised per patient.
- Drawbacks: requires manufacturing lead time; cannot be modified intraoperatively (anatomy seen intraoperatively may differ from the preoperative model); anatomy-dependent (a different template is needed for each vertebra and each patient).

Patient-mounted robot (MARS, Mazor Surgical Technologies, Israel): A miniature 6-DOF parallel robot (Stewart platform variant) is clamped rigidly onto the vertebra. Because it moves with the bone, any global patient motion (breathing, table repositioning) is automatically compensated — the robot's frame of reference moves with the patient. It positions a drill guide to sub-millimetre accuracy; the surgeon drills manually through the guide.

> **The economic dimension.** One of the key factors in the adoption of a medical robot by a hospital is not the absolute cost, but the *cost-benefit ratio*: how many patients can be treated, how many complications prevented, how much OR time saved. The MARS robot, despite its technical cleverness, was anatomy-specific (designed for spine surgery only) and had a limited range of application. In evaluating economic impact, this limited versatility weakened the case for adoption. The contrast with the template — which is cheaper but requires manufacturing time and cannot adapt intraoperatively — illustrates that the choice between robotic and non-robotic solutions is always an economic and practical trade-off, not just a technical one.

5. Classification 4: Dario (2004) — 2D Matrix

Paolo Dario (formerly of Scuola Superiore Sant'Anna, Pisa) proposed a **two-dimensional classification matrix** combining:

- **Type of interaction** (rows): Autonomous, Interactive, Teleoperated, Passive.
- **Type of surgical access** (columns): Traditional/open, Minimally invasive, Endocavitary/endoluminal.

	Traditional access	**MIS access**	**Endoluminal access**
Autonomous	ROBODOC, CASPAR	Stereotaxis Inc.	MUSYC / EMIL
Interactive	Eye scalpel, RinC	AESOP, MIAS	Active catheters
Teleoperated	Mammotome, PAKY	Da Vinci, Zeus	MiNOSC
Passive	PinPoint	HALS (not robotic)	Given Imaging (not robotic)

The importance of this matrix is that it acknowledges a fact the Troccaz classification obscures: **the type of access fundamentally changes the engineering requirements**. An autonomous system operating on open bone (ROBODOC) and an autonomous system navigating inside the GI tract (an active capsule) are both "autonomous" but have almost nothing else in common in terms of kinematics, actuation, sensing, and control. Grouping them under a single label without the second axis of surgical access loses critical information.

Note also that the Dario matrix dates from 2004, so some entries were research prototypes rather than clinical systems. The endoluminal autonomous column, largely empty then, is being populated now by devices such as the Ion bronchoscope and steerable capsules.

> **Empty cells as research opportunities.** Any empty cell in the Dario matrix represents either: (a) a clinically meaningful combination that is technically unsolved — a research opportunity — or (b) a combination that has no clinical relevance. Analysing *why* a cell is empty is as informative as the filled cells themselves.

6. The Regulator's Viewpoint: IEC 80601-2-77 (2019)

The regulatory perspective on classification differs from the engineering perspective. For engineers seeking certification of a new surgical robot, the relevant standard is IEC 80601-2-77 (2019), which provides particular requirements for the basic safety and essential performance of *robotically assisted surgical equipment*. It defines:

> **Robotically Assisted Surgical Equipment (RASE):** Medical electrical equipment incorporating a Programmable Electrical Medical System (PEMS) actuated mechanism intended to facilitate the placement or manipulation of a robotic surgical instrument.
>
> **Robotic surgical instrument:** An invasive device with an applied part, intended to be manipulated by RASE to perform tasks in surgery.

There is currently no regulatory framework written *specifically* for surgical robots. The applicable framework is a composite of medical device regulations and service/industrial robotics standards. This standard is not a law but a set of rules: demonstrating compliance facilitates and accelerates the certification process. Understanding the certification pathway is essential for any engineer designing a surgical robot, because the timeline from prototype to clinical approval is 7–10 years minimum.

Classification of surgical invasiveness for the purposes of this standard:

Invasiveness type	**Examples**
Energy only through body surface	Radiosurgery (CyberKnife), focused ultrasound, shoulder relocation
Physical invasion through body surface	Open surgery, bone fixation, bone milling, laparoscopic surgery, vascular intervention
Energy via natural orifice	Focused ultrasound via endoscope
Physical invasion via natural orifice	Endomucosal resection, dental milling

Essential performance. The standard requires demonstrating that the device maintains its *essential performance* — the minimum functional requirements for patient safety — under fault conditions. For an autonomous bone-milling robot, essential performance includes the ability to detect and stop if the tool exits the planned region. For a teleoperated system, it includes maintaining positional accuracy under communication delays. The classification determines which essential performance requirements apply. Engineers must identify these requirements at the *beginning* of the design process, not as an afterthought.

7. Modern Classification: Level of Autonomy (LoA)

As AI and autonomy become increasingly relevant in surgical robotics, classification based on **Level of Autonomy (LoA)** provides the most forward-looking framework. It is directly analogous to the SAE levels of driving automation and was formalised for surgery by Haidegger (2019) and extended by Lee et al. (2024) with a decision-tree approach applicable to FDA-cleared devices.

7.1. The LoA Scale (Haidegger, 2019)

LoA	Name	Description
0	No autonomy	Robotic system identical to manual surgery. No active role for the system.
1	Robot assistance	Surgeon directly and continuously controls all robot motion. System provides passive support (tremor filtering) and/or active guidance (haptic feedback). Telerobotic capabilities.
2	Task-level autonomy	Surgeon provides task parameters; system takes over control to execute the task. Controls not released during autonomous execution.
3	Conditional / supervised autonomy	System independently generates potential strategies for surgeon to select. Large sections executed autonomously; system may prompt for safe handover.
4	High-level autonomy	System selects the optimal patient-specific plan and executes autonomously. Surgeon only approves (e.g., radiotherapy from a computer-generated plan).
5	Full autonomy	System independently decides and executes the entire procedure. No fallback. Does not exist clinically.

Mapping known systems to the LoA scale. Da Vinci (standard operation): LoA 1 — surgeon continuously controls all motions; the system provides tremor filtering and scaling but makes no independent decisions. ROBODOC / TSolution One during autonomous milling: LoA 2 — surgeon has provided the plan; robot executes while surgeon supervises. CyberKnife during beam delivery with real-time motion compensation: approaching LoA 4 — the system autonomously adapts beam direction based on continuous imaging; the surgeon's role is supervisory. No currently deployed surgical system operates at LoA 5.

7.2. Decision-Tree Framework for LoA (Lee et al., 2024)

Lee et al. provide a practical decision-tree for systematically determining the LoA of any FDA-cleared surgical robot:

1. Does the surgeon continuously control all system motions?
   - Yes → Does the system aid execution/monitoring? → Yes: **LoA 1**; No: **LoA 0**.
   - No → proceed.
2. Does the system independently generate and update patient-specific strategies?
   - No → **LoA 2**: system executes a surgeon-parameterised task.
   - Yes → proceed.
3. Does the system independently select the optimal strategy?
   - No → **LoA 3**: system proposes strategies; surgeon selects; system executes.
   - Yes → proceed.
4. Does the surgeon need to approve the plan?
   - Yes → **LoA 4**: system selects and executes; surgeon approves.
   - No → **LoA 5**: fully autonomous.

7.3. Examples of FDA-Cleared Systems by LoA

Device	Specialty	LoA	Key features	AI?
Senhance (Asensus)	Gen. Surg	2	Anatomical identification, haptic feedback, eye-tracking	Yes
CorPath GRX (Corindus)	Interventional	2	Automated catheter navigation; remote joystick	No
AquaBeam (Procept)	Urology	2	Ultrasound-planned waterjet tissue removal	No
TMINI (Think Surgical)	Orthopaedics	2	CT-planned; hand-held tool auto-compensates motion	No
Mona Lisa 2.0 (Biobot)	Urology	3	System proposes biopsy plan; auto-positions needle guide	No
TSolution One (Think)	Orthopaedics	3	Planning software designs procedure; active milling under supervision	No
ARTAS iX (Venus)	Hair restoration	3	AI maps each follicle; learns graft patterns; image-guided harvesting	Yes

8. Future Trends: AI-Enhanced Surgical Systems

AI is gradually transforming surgical robotics across the full workflow: preoperative planning, intraoperative guidance, performance evaluation, and autonomous robotic action.

8.1. What Is Already Happening

Preoperative planning: AI tools for image classification, detection, segmentation, and registration are in clinical use. Challenges include multi-modal input, patient variability, and limited annotated training data.

Intraoperative guidance: real-time 3D shape instantiation, endoscopic scene navigation, tissue feature tracking, and augmented reality overlays. Challenges: limited-resolution intraoperative imaging; tissue deformation between the preoperative model and the intraoperative reality.

Surgical scene understanding: AI systems that automatically recognise what is happening in the surgical field — which tool is doing what, whether a tool is in contact with tissue it should not touch, whether the procedure is progressing normally. This is important for two reasons:

- **Error avoidance**: for example, in laparoscopic cholecystectomy (the most common surgical procedure performed worldwide), there exists a specific camera viewpoint configuration that is known to be associated with bile duct injury — a serious complication. AI systems trained to recognise this dangerous configuration can alert the surgeon before an error occurs.
- **Trainee assessment**: automatically evaluating the performance of a surgical trainee requires understanding the scene — how many times did a tool make inadvertent contact with tissue, how long did tissue retraction take, was a particular anatomical structure correctly identified. Expert surgeons must annotate training videos to create the labelled datasets required for this, which is expensive and slow.

> **The data annotation challenge.** Most surgeries worldwide are still performed manually, not robotically. The data available for training AI models is therefore predominantly 2D video from standard laparoscopic cameras, not kinematic data from robots. Extracting information from 2D video — whether a tool is in contact with a specific tissue type, what force is being applied, what anatomical structure is visible — is challenging and requires large volumes of labelled data. Expert surgeons must annotate those videos, identifying every relevant event. The cost of creating such annotated datasets is a significant barrier to progress in AI-based surgical performance assessment.
>
> **The interplay between robotics and AI.** When a procedure is performed robotically, localisation of the tools is much easier: the robot's own sensors (encoders, force sensors) provide kinematic data directly, and the tool position is known in the robot frame. Once the robot frame is registered to the preoperative plan, the tool position relative to the planned anatomy is known continuously. Moreover, in surgical robotic systems vision is often 3D, thus providing depth information. This makes AI-based scene understanding and performance assessment much easier in robotic surgery than in manual surgery — which is another argument for increasing the proportion of procedures performed robotically, beyond the direct clinical benefit to the individual patient.

8.2. What Will Happen

Near-future developments identified in the literature:

Domain	Expected development
Preoperative planning	Large-scale datasets; federated learning and meta-learning for explainable AI; early detection from multimodal data
Intraoperative guidance	Dynamic organ function displays; advanced AR/VR for training; remote multi-disciplinary surgical cooperation
Surgical robotics	More versatile, lighter, cheaper robots; increased LoA via learning from demonstration (LfD) and reinforcement learning (RL); nanorobots for diagnosis and drug delivery
Ethical/legal	Patient data privacy; cybersecurity for AI-enabled systems; liability frameworks; trust between patients and AI systems

Learning from demonstration (LfD) and reinforcement learning (RL). These are the two main approaches being explored for increasing surgical robot autonomy. In LfD, the robot learns a surgical skill from kinematic and video recordings of expert surgeons. In RL, it learns by trial and error in simulation, guided by a reward signal. LfD is more data-efficient but limited to replicating observed behaviour; RL can potentially discover novel strategies but requires accurate simulation and careful reward design. Both face the fundamental challenge of *generalisation*: a skill learned on one patient anatomy must transfer to different anatomies, tissue states, and intraoperative situations.

9. Comparison of Classification Schemes

Author / Year	Perspective	Classification axes	Strength
Troccaz 2001	Robotics & control	Interaction mode (passive / active / interactive / teleoperated)	Simple; maps cleanly to hardware and control architecture
Taylor 2003	CIS workflow	Role in CIS (CAD/CAM vs surgical assistant)	Integrates planning, execution, assessment; system-level view
Joskowicz 2005	Clinical	Support device type (template → navigation → robot; floor/patient mount)	Clinical perspective; captures non-robotic solutions
Dario 2004	Systems engineering	2D matrix: interaction × access type	Systematic; maps systems to unique cell; exposes gaps
Haidegger 2019	Autonomy & AI	LoA 0–5 (analogous to SAE vehicle levels)	Future-proof; handles AI-enabled systems
Lee et al. 2024	Regulatory	Decision-tree LoA for FDA-cleared devices; AI/ML flag	Data-driven; grounded in clinical evidence

10. The Course Framework: Classification by Robotic Task

The framework used throughout this course classifies surgical systems by **domain of use and associated robotic task**. This is chosen because it is the most useful lens for the engineering analysis of design methodologies: it directly connects the clinical problem to the specific kinematics, control modality, sensors, and actuators required. It corresponds roughly to the rows of the Dario matrix (access type) combined with the Troccaz interaction mode, but at a finer level of granularity.

Domain	Robotic task	Kinematics	Control	Sensors / Actuators
Orthopedics	Machining of rigid surface	Conventional serial/parallel + wrist; 5–6 DOF	Autonomous; cooperative; hands-on	Conventional; vel: mm/s; force: ≤100 N
MIS	Constrained manipulation (RCM)	Passive RCM; 5–6 ext. + extra int. DOF	Teleoperation; shared	Conventional; vel: cm/s; force: few N
Neurosurgery / IR / radiotherapy	Constrained targeting	Conventional + dedicated front-end; 5–6 DOF	Teleoperation; (semi-)autonomous	MR/CT compatible (pneumatic, ultrasonic); force: few N
Microsurgery	Micromanipulation	Dedicated kinematic architecture; possible RCM	Shared / cooperative teleoperation	Piezo, ultrasonic; force: few mN
Tele-echo / TMS	Surface tracking	Conventional + dedicated wrist; possible RCM	Autonomous; teleoperation; shared	Conventional; vel: mm/s; force: few N

11. Summary

- Classification is a tool, not an end. Each scheme illuminates different aspects of the design space and all are complementary.
- **Troccaz (2001)**: passive / active / interactive (semi-active, synergistic) / teleoperated. Maps to hardware and control architecture. Introduced the concept of shared control in surgery.
- **Taylor (2003)**: surgical CAD/CAM vs. surgical assistants (extenders and auxiliary supports), within the CIS framework. Key insight: a surgical robot is never a standalone device; it is part of an information-processing system spanning planning, execution, and assessment.
- **Joskowicz (2005)**: templates → navigation → floor/bed robots → patient-mounted robots. Clinical perspective; includes non-robotic alternatives. Economic factors are as important as technical performance in adoption.
- **Dario (2004)**: 2D matrix (interaction × access). Exposes gaps and shows that "autonomous" means very different things depending on the access modality.
- **LoA scale (Haidegger 2019; Lee et al. 2024)**: LoA 0–5. Commercial systems are at LoA 1–3; LoA 4 exists for specific applications; LoA 5 does not exist clinically. AI will drive progressive increases in LoA.
- **IEC 80601-2-77**: defines RASE and essential performance by invasiveness type. Essential

for the certification pathway.

Key topics for revision.

1. Troccaz's four categories with one clinical example each.
2. Difference between semi-active (mechanical constraint, NeuroMate) and synergistic (programmable constraint, PADyC / Acrobot). Why was PADyC's approach technically challenging?
3. PADyC's freewheel mechanism: modes F_1–F_4; the WAC and its computation. Why is mapping task-space constraints to joint space non-trivial?
4. Acrobot / MAKO: virtual fixture via impedance control. Transparent inside the safe zone; resistive at the boundary.
5. Taylor's CIS framework: the three phases; the role of registration as the "feedback" element; surgical CAD/CAM vs. surgical assistants.
6. Joskowicz's classification: the template–navigation–robot spectrum; economic factors in adoption.
7. Dario's 2D matrix: what the axes are; why empty cells matter; where Da Vinci and ROBODOC sit.
8. LoA 0–5: description of each level; placement of Da Vinci (LoA 1), ROBODOC (LoA 2), CyberKnife (LoA 4).
9. Lee et al. decision tree: the four key questions and their branching.
10. AI trends: what is happening now (scene understanding, trainee assessment) vs. what will happen (LfD, RL, nano-robots). The annotation bottleneck.

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